管理SVM中的路由

ONTAP 9

NetApp
May 03, 2022

This PDF was generated from https://docs.netapp.com/zh-tw/ontap/networking/manage_routing_in_an_svm_overview.html on May 03, 2022. Always check docs.netapp.com for the latest.

目錄

管理SVM中的路由 1
總覽 1
建立靜態路由 1
啟用多重路徑路由 1
刪除靜態路由 2
顯示路由資訊 2
從路由表中移除動態路由 4

管理SVM中的路由

總覽

SVM的路由表會決定SVM用來與目的地通訊的網路路徑。瞭解路由表的運作方式非常重要、如此一來、您就能在網路問題發生之前先行防範。

路由規則如下：

- 透過最具體的可用路由傳送流量。ONTAP
- 當無法使用更多特定路由時、透過預設閘道路由（網路遮罩為0位元）路由流量。ONTAP

如果路由具有相同目的地、網路遮罩和度量、則無法保證系統在重新開機或升級後會使用相同的路由。如果您已設定多個預設路由、這尤其會造成問題。

最佳做法是僅為SVM設定一條預設路由。為避免中斷、您應確保預設路由能夠到達其他特定路由無法到達的任何網路位址。如需詳細資訊、請參閱知識庫文章 "SU134：叢集ONTAP 式的故障路由組態不正確、可能會中斷網路存取"

建立靜態路由

您可以在儲存虛擬機器（SVM）內建立靜態路由、以控制LIF如何將網路用於傳出流量。

當您建立與SVM相關聯的路由項目時、該路由會被指定SVM擁有且與閘道位於同一子網路的所有LIF使用。

使用「network route create」命令建立路由。

```
network route create -vserver vs0 -destination 0.0.0.0/0 -gateway
10.61.208.1
```

啟用多重路徑路由

如果多個路由的目的地度量相同、則只會針對傳出流量選取其中一個路由。這會導致傳送傳出流量的其他路由未使用。您可以啟用多重路徑路由來平衡負載、並使用所有可用的路由。

步驟

1. 登入進階權限層級：

   "進階權限"

2. 啟用多重路徑路由：

   「network options multipath-ro路 由修改-is啟用true」

   叢集中的所有節點均已啟用多重路徑路由。

```
network options multipath-routing modify -is-enabled true
```

刪除靜態路由

您可以從儲存虛擬機器（SVM）刪除不需要的靜態路由。

使用「network route DELETE」命令刪除靜態路由。

有關此命令的更多信息，請參見"network route"手冊頁： "指令ONTAP"。

下列範例會刪除與SVM vs0關聯的靜態路由、閘道為10.63.0.1、目的地IP位址為0.00.0/0：

```
network route delete -vserver vs0 -gateway 10.63.0.1 -destination
0.0.0.0/0
```

顯示路由資訊

您可以顯示叢集上每個SVM的路由組態資訊。這有助於診斷與用戶端應用程式或服務之間的連線問題有關的路由問題、以及叢集中某個節點上的LIF。

步驟

1. 使用「network routes show」命令來顯示一或多個SVM內的路由。下列範例顯示在vs0 SVM中設定的路由：

   ```
   network route show
     (network route show)
   Vserver             Destination     Gateway         Metric
   ------------------- --------------- --------------- ------
   vs0
                       0.0.0.0/0       172.17.178.1    20
   ```

2. 使用「network route show-lifs」命令、顯示一或多個SVM中路由和lifs的關聯。

   以下範例顯示vs0 SVM擁有路由的lifs：

```
network route show-lifs
  (network route show-lifs)

Vserver: vs0
Destination             Gateway                 Logical Interfaces
----------------------  --------------          --------------------
0.0.0.0/0               172.17.178.1            cluster_mgmt,
                                                LIF-b-01_mgmt1,
                                                LIF-b-02_mgmt1
```

3. 使用「network route active-entry show」命令、可在一個或多個節點、SVM、子網路或具有指定目的地的路由上顯示已安裝的路由。

   下列範例顯示特定SVM上所有已安裝的路由：

```
network route active-entry show -vserver Data0

Vserver: Data0
Node: node-1
Subnet Group: 0.0.0.0/0
Destination             Gateway              Interface   Metric  Flags
----------------------  -------------------  ---------   ------  -----
127.0.0.1               127.0.0.1            lo              10  UHS
127.0.10.1              127.0.20.1           losk            10  UHS
127.0.20.1              127.0.20.1           losk            10  UHS

Vserver: Data0
Node: node-1
Subnet Group: fd20:8b1e:b255:814e::/64
Destination             Gateway              Interface   Metric  Flags
----------------------  -------------------  ---------   ------  -----
default                 fd20:8b1e:b255:814e::1
                                             e0d             20  UGS
fd20:8b1e:b255:814e::/64
                        link#4               e0d              0  UC

Vserver: Data0
Node: node-2
Subnet Group: 0.0.0.0/0
Destination             Gateway              Interface   Metric  Flags
----------------------  -------------------  ---------   ------  -----
127.0.0.1               127.0.0.1            lo              10  UHS

Vserver: Data0
Node: node-2
```

```
Subnet Group: 0.0.0.0/0
Destination             Gateway              Interface   Metric  Flags
----------------------  -------------------  ---------   ------  -----
127.0.10.1              127.0.20.1           losk            10  UHS
127.0.20.1              127.0.20.1           losk            10  UHS

Vserver: Data0
Node: node-2
Subnet Group: fd20:8b1e:b255:814e::/64
Destination             Gateway              Interface   Metric  Flags
----------------------  -------------------  ---------   ------  -----
default                 fd20:8b1e:b255:814e::1
                                             e0d             20  UGS
fd20:8b1e:b255:814e::/64
                        link#4               e0d              0  UC
fd20:8b1e:b255:814e::1  link#4               e0d              0  UHL
11 entries were displayed.
```

從路由表中移除動態路由

收到針對IPv6和IPv6的ICMP重新導向時、動態路由會新增至路由表。根據預設、動態路由會在300秒後移除。如果您想要在不同時間內維持動態路由、可以變更逾時值。

您可以將逾時值從0設定為65,535秒。如果您將值設為0、則路由永遠不會過期。移除動態路由可避免因無效路由持續存在而導致連線中斷。

步驟

1. 顯示目前的逾時值。
   - 對於IPV4：

     ```
     network tuning icmp show
     ```

   - 對於IPv6：

     ```
     network tuning icmp6 show
     ```

2. 修改逾時值。
   - 對於IPV4：

     ```
     network tuning icmp modify -node node_name -redirect-timeout
     timeout_value
     ```

- 對於IPv6：

```
network tuning icmp6 modify -node node_name -redirect-v6-timeout
timeout_value
```

3. 確認逾時值已正確修改。
   - 對於IPV4：

```
network tuning icmp show
```

   - 對於IPv6：

```
network tuning icmp6 show
```

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.